

Periodic Anchors and Exact Mixed-Moment Phase Diagrams

Singular coboundaries, explicit freezing surfaces,
and arithmetic leading amplitudes

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Abstract

We identify a globally solvable class of genuinely mixed digital products. For the map $T_b x = bx \pmod{1}$, place phase shifts on finitely many complete periodic orbits $\mathcal{O}_i$, with a common positive exponent a_i on each orbit. The product of normalized digital masks is a singular coboundary. It changes the energy of every invariant measure avoiding the selected orbits by the same amount, but gives each selected orbit an explicit additional reward. For an arbitrary continuous nonnegative background weight v of finite pressure, we prove

$$\mathcal{P}_b \left(v \prod_i \prod_{p \in \mathcal{O}_i} m_b(\cdot - p)^{a_i} \right) = -D \log b + \max \left\{ \mathcal{P}_b(v), \max_i (Q_i(v) + a_i \log b) \right\},$$

where $D = \sum_i a_i |\mathcal{O}_i|$ and $Q_i(v)$ is the orbit average of $\log v$. For a squared shifted digital background, Parseval gives $\mathcal{P}_b(v) = 0$ at every phase. A subharmonic sampling estimate converts this formula into explicit mixed-moment growth exponents for arbitrary positive real anchor exponents. Without a background we also determine leading constants, logarithmic critical terms, and uniform critical windows. An exactly evaluated seventh-root example has leading amplitudes 3, 4, 4 according to the depth modulo three. We construct prescribed finite polyhedral pressure diagrams and extend orbit selection to nonlinear expanding circle maps. Full proofs, exact-arithmetic checks, numerical diagnostics, and a research agenda are included.

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1 Research target, contribution, and limits

1.1 Why mixed digital products are a useful next target

The Fabius–Rvachev–Thue–Morse part of ProveIt contains finite-product identities, moment computations, and recent manuscripts on integer, fractional, critical, and subcritical digital-product pressure [5, 6]. The single-phase critical law is already treated there. This article therefore does not claim to discover that law or to fill an endpoint gap already addressed by later drafts. It changes the problem: several phase-shifted digital products are multiplied before taking a moment.

The recent primary literature relates generalized Thue–Morse Riesz products to thermodynamic pressure and fractal spectra [2]. Continuity in the location of a logarithmic singularity, and the problem of stronger positive-order phase regularity, are studied in [3]. Our results concern an exactly structured mixed family, not a general solution of that phase-regularity problem.

For a single atomic mask, telescoping is familiar. The additional observation is that telescoping survives for an entire periodic orbit of phase shifts, and for any finite collection of such orbits. The singularity then distinguishes measures supported on the selected orbits from all other invariant measures. This yields a finite maximum of explicit pressure branches rather than a perturbative approximation to an unknown spectral radius.

The principal payoff is the exact mixed-moment exponent in theorem 2.3. It holds globally in the background phase, in every integer base, and for arbitrary positive real anchor exponents. The orbit-balance condition is essential. We make no claim to evaluate arbitrary unrelated phase tuples.

Editorial note (ProveIt, 2026-09-30). Four research questions of the pressure articles beside this one, in Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/thue-morse/, concern related situations: “Nonatomic phases” of Thue_Morse_Critical_Pressure/, “Regularity at nonzero phases” and “Other masks and multiple zeros” of Thue_Morse_Fractional_Pressure/, and “Beyond one zero and beyond base two” of Thue_Morse_Integer_Pressure/. This article bears on them without answering any. It evaluates exactly the pressure of products of shifted masks whose phases fill complete periodic orbits with a common exponent on each orbit (theorem 2.1), at fixed phases; a single mask whose phase has period at least two is not orbit-balanced and is not covered, and no regularity in the phase is proved. Each of those questions now carries a note pointing here. With the single anchor 0, corollary 2.2 gives $B(\max\{1, a\} - a)$, which agrees with the atomic-phase pressure values of the subcritical and fractional pressure articles (checked on filing).

1.2 What is proved, and what is inherited

A one-sided integrability lemma makes singular coboundary cancellation legitimate. A variational principle with zero weights turns the resulting energy identity into an exact pressure formula and a classification of its equilibrium faces. A sampling estimate for products of absolute powers of polynomials identifies the corresponding moment exponent. The thermodynamic inputs inherited from classical theory are the continuous-potential variational principle and the usual entropy facts for an expanding map [1]; the extension at zeros is proved here.

No fractional-cusp, critical-cusp, or spectral-perturbation theorem from an unreviewed repository draft is needed in this chain. The exact finite-product normalization is also derived directly. Future revisions of the neighboring pressure manuscripts therefore do not affect these proofs.

For orbit-balanced products without a background, we determine subcritical constants, critical logarithms, supercritical residue-class amplitudes, and a multivariate critical crossover. An explicit three-phase example is evaluated exactly at every depth, not merely to leading order.

1.3 Provenance and novelty boundary

The repository index was inspected at the pinned commit

781594d886f8dc56069221e2b56bd1e94d9c6e5f.

The relevant index is under

Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/thue-morse/README.md.

It explicitly labels the recent pressure arrivals unreviewed and without Lean statements. We retain the distinction between mathematical exposition, numerical evidence, and formal proof.

The finite polyhedral realization result in section 11 is not a claim that pressure flexibility is new. Kucherenko and Quas prove much broader realization results for continuous potentials on compact symbolic systems [4]. Our contribution there is a finite explicit trigonometric construction on a fixed expanding circle map, with chosen periodic equilibrium measures and logarithmically singular potentials. The restricted source search did not establish an independent priority claim for the complete package of formulas below.

2 Definitions and principal formulas

Fix an integer $b \geq 2$, put $B = \log b$, and write $T = T_b: \mathbb{T} \rightarrow \mathbb{T}$, $Tx = bx \pmod{1}$, where $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. All logarithms are natural unless a base is displayed. Define

$$d_b(x) = \frac{1}{b} \sum_{r=0}^{b-1} e^{2\pi i r x}, \quad m_b(x) = |d_b(x)| = \left| \frac{\sin \pi b x}{b \sin \pi x} \right|. \quad (2.1)$$

At integers the quotient has removable value 1. Thus $0 \leq m_b \leq 1$, and $m_2(x) = |\cos \pi x|$.

2.1 Pressure for a weight which may vanish

For a continuous $w: \mathbb{T} \rightarrow [0, \infty)$ set

$$w^{(n)}(x) = \prod_{j=0}^{n-1} w(T^j x), \quad N = b^n, \quad (2.2)$$

$$Z_n(w) = \sum_{k=0}^{N-1} \sup_{x \in [k/N, (k+1)/N]} w^{(n)}(x), \quad \mathcal{P}_b(w) = \lim_{n \rightarrow \infty} \frac{1}{n} \log Z_n(w). \quad (2.3)$$

The limit and its variational meaning are proved in section 3. We write $\mathcal{P}_b(w)$ rather than $P(\log w)$ to keep zeros visible. Thus $\mathcal{P}_b(cw) = \log c + \mathcal{P}_b(w)$ for $c > 0$ and $\mathcal{P}_b(1) = B$. An equilibrium measure maximizes $h_\mu(T) + \int \log w \, d\mu$, with $\log 0 = -\infty$.

There is no factor $1/b$ in the associated transfer operator $L_w f(x) = \sum_{Ty=x} w(y)f(y)$. The exponent of a Lebesgue moment is $\mathcal{P}_b(w) - B$, not $\mathcal{P}_b(w)$, whenever the moment-pressure conversion below applies.

2.2 Periodic anchors

Let $\mathcal{O}_1, \dots, \mathcal{O}_m$ be pairwise disjoint periodic orbits of T , with lengths r_i . Give every point of $\mathcal{O}_i$ the same exponent $a_i > 0$. Put

$$E = \bigcup_i \mathcal{O}_i, \quad D = \sum_i r_i a_i, \quad (2.4)$$

$$A_{\mathbf{a}}(x) = \prod_i \prod_{p \in \mathcal{O}_i} m_b(x - p)^{a_i}, \quad \Phi_{\mathbf{a}}(x) = \prod_i \prod_{p \in \mathcal{O}_i} |\sin \pi(x - p)|^{a_i}. \quad (2.5)$$

Exponent zero means that the orbit is omitted; no interpretation of 0^0 is needed. For a background $v \geq 0$ define

$$Q_i(v) = \frac{1}{r_i} \sum_{p \in \mathcal{O}_i} \log v(p). \quad (2.6)$$

If v vanishes at one point of $\mathcal{O}_i$, then $Q_i(v) = -\infty$. Let $\delta_i = r_i^{-1} \sum_{p \in \mathcal{O}_i} \delta_p$.

Theorem 2.1 (Exact periodic-anchor pressure). *Let $v: \mathbb{T} \rightarrow [0, \infty)$ be continuous with $\mathcal{P}_b(v) > -\infty$. Then*

$$\mathcal{P}_b(vA_{\mathbf{a}}) = -DB + \max \left\{ \mathcal{P}_b(v), \max_i (Q_i(v) + a_i B) \right\}. \quad (2.7)$$

Write the maximum as C . The equilibrium measures of $vA_{\mathbf{a}}$ are exactly the convex combinations of the following available measures: all equilibrium measures of v when $C = \mathcal{P}_b(v)$, and δ_i for each i with $Q_i(v) + a_i B = C$.

Whenever the background branch wins, all its equilibrium measures give zero mass to E . No uniqueness hypothesis on the background is required.

Corollary 2.2 (No background). *For $v = 1$,*

$$\mathcal{P}_b(A_{\mathbf{a}}) = B (\max\{1, a_1, \dots, a_m\} - D). \quad (2.8)$$

If all $a_i < 1$, the unique equilibrium measure is Lebesgue measure. If the largest $a_i > 1$, equilibrium measures are exactly the mixtures of orbit measures with that largest exponent. At a maximum equal to 1, Lebesgue measure and precisely the orbits with exponent 1 coexist.

The maximal-entropy measure for T_b is uniquely Lebesgue measure: the uniform Bernoulli coding has entropy $\log b$, and equality in the finite-alphabet entropy bound forces independent uniform digits.

2.3 The mixed digital moment

Let $s_b(k)$ denote the sum of the base- b digits of k , and put

$$S_{n,c}(x) = \sum_{k=0}^{b^n-1} e^{2\pi i(kx - cs_b(k))}. \quad (2.9)$$

This sign convention gives binary classical Thue–Morse phase $c = 1/2$ and Dirichlet-kernel phase $c = 0$. For a squared background set

$$Q_i(c) = \frac{2}{r_i} \sum_{p \in \mathcal{O}_i} \log m_b(p - c). \quad (2.10)$$

Theorem 2.3 (Global mixed-moment exponent). *For every real phase c and every positive real vector $\mathbf{a}$, the moment*

$$I_n(\mathbf{a}, c) = \int_0^1 |S_{n,c}(x)|^2 \prod_i \prod_{p \in \mathcal{O}_i} |S_{n,p}(x)|^{a_i} dx \quad (2.11)$$

satisfies, with $N = b^n$,

$$\lim_{n \rightarrow \infty} \frac{\log I_n(\mathbf{a}, c)}{\log N} = 1 + \max \left\{ 0, \max_i \left(a_i + \frac{Q_i(c)}{B} \right) \right\}. \quad (2.12)$$

Branches with $Q_i(c) = -\infty$ are omitted. Equivalently, if the exponent is γ , then $I_n = N^{\gamma+o(1)}$.

This is not an assertion that I_n/N^γ converges. Logarithmic factors and arithmetic oscillations are not ruled out. Both phenomena will occur in the family without a background.

3 Two foundations: zero weights and singular cancellation

3.1 A variational principle which includes zeros

Lemma 3.1. *For every continuous $w \geq 0$, the limit in equation (2.3) exists in $[-\infty, \infty)$ and*

$$\mathcal{P}_b(w) = \sup_{\mu \in \mathcal{M}_T} \left(h_\mu(T) + \int \log w \, d\mu \right), \quad (3.1)$$

where $\mathcal{M}_T$ is the set of invariant Borel probability measures. If $\mathcal{P}_b(w) > -\infty$, the supremum is attained.

Proof. Every $(n+k)$ -cylinder is specified by an n -cylinder and a k -cylinder for its image under T^n . Splitting the product at time n gives $Z_{n+k}(w) \leq Z_n(w)Z_k(w)$. Fekete's lemma yields $\mathcal{P}_b(w) = \inf_{n \geq 1} n^{-1} \log Z_n(w)$. If some $Z_n = 0$, later products vanish and the extended-real interpretation is immediate.

For $\varepsilon > 0$, the potential $\log(w + \varepsilon)$ is continuous, so the ordinary variational principle [1] applies. The b -adic coding identifies our partition sum with the full-shift cylinder sum. The coding is at most two-to-one and adds zero relative entropy; it preserves the variational value. Thus this application has exactly our normalization.

For fixed n , uniform convergence gives $Z_n(w + \varepsilon) \rightarrow Z_n(w)$. Consequently

$$\mathcal{P}_b(w + \varepsilon) \downarrow \mathcal{P}_b(w). \quad (3.2)$$

Monotonicity gives one bound; for each fixed n the limiting pressure is at most $n^{-1} \log Z_n(w)$, giving the other after taking the infimum.

The entropy map is upper semicontinuous for T_b , by positive expansiveness. To justify the singular limit explicitly, choose equilibrium measures μ_j for $w + \varepsilon_j$, with $\varepsilon_j \downarrow 0$, and take a weakly convergent subsequence $\mu_j \rightarrow \mu$. For fixed $\delta > 0$ and large j ,

$$\mathcal{P}_b(w + \varepsilon_j) \leq h_{\mu_j}(T) + \int \log(w + \delta) \, d\mu_j.$$

Upper semicontinuity of entropy and continuity of the second term give

$$\mathcal{P}_b(w) \leq h_\mu(T) + \int \log(w + \delta) \, d\mu.$$

Let $\delta \downarrow 0$, using monotone convergence after subtracting an upper bound. Conversely, apply the continuous variational principle to any invariant measure and take the same limit. This proves equation (3.1); when the value is finite the limiting measure attains it. If the value is $-\infty$, the converse bound already forces every variational value to be $-\infty$. $\square$

3.2 Why infinite logarithmic integrals cannot simply be canceled

A formal identity $g = u \circ T - u$ does not permit subtraction of two infinite integrals. The following lemma is the precise substitute.

Lemma 3.2 (One-sided stationary cancellation). *Suppose T preserves a probability measure μ and $U: X \rightarrow [0, \infty)$ is finite almost everywhere. If $U - U \circ T$ is bounded above, then it is integrable and*

$$\int (U - U \circ T) \, d\mu = 0. \quad (3.3)$$

The same holds with “bounded below” in place of “bounded above.”

Proof. Let $U_M = \min(U, M)$ and $d_M = U_M - U_M \circ T$. Stationarity gives $\int d_M = 0$. For finite $u, v \geq 0$, each of

$$(\min(u, M) - \min(v, M))^+, \quad (\min(u, M) - \min(v, M))^-$$

increases to the corresponding part of $u - v$. If $d = U - U \circ T \leq C$, then $d_M^+ \leq d^+ \leq \max(C, 0)$. Thus $\int d_M^+$ converges to the finite integral of d^+ . Since $\int d_M^- = \int d_M^+$, monotone convergence gives the same finite integral for d^- . Their difference is zero. Interchange the parts to treat a lower bound. Integrability of U was not assumed. $\square$

The bound needed here is automatic for the logarithm of a continuous nonnegative weight: it is bounded above, though it may be unbounded below.

4 The orbit coboundary and the exact pressure theorem

4.1 An exact finite identity

Lemma 4.1 (Orbit-balanced coboundary). *For $x \notin E$,*

$$A_{\mathbf{a}}(x) = b^{-D} \frac{\Phi_{\mathbf{a}}(Tx)}{\Phi_{\mathbf{a}}(x)}. \quad (4.1)$$

If $p \in \mathcal{O}_i$, define

$$c_p = \pi^{a_i} \prod_{q \in E \setminus \{p\}} |\sin \pi(p - q)|^{a(q)}, \quad a(q) = a_j \quad (q \in \mathcal{O}_j). \quad (4.2)$$

Then $c_p > 0$ and the continuous value at p is

$$A_{\mathbf{a}}(p) = b^{a_i - D} \frac{c_{Tp}}{c_p}. \quad (4.3)$$

Proof. Multiplying equation (2.1) over an orbit gives numerator $\prod_{p \in \mathcal{O}_i} |\sin \pi(bx - bp)|^{a_i}$. Since $p \mapsto bp \pmod{1}$ permutes $\mathcal{O}_i$, this equals $\prod_{p \in \mathcal{O}_i} |\sin \pi(bx - p)|^{a_i}$. Multiply over i . Near $p \in \mathcal{O}_i$, all factors except the one at p are smooth and strictly positive, so

$$\Phi_{\mathbf{a}}(p + y) = c_p |y|^{a_i} (1 + O(|y|)). \quad (4.4)$$

The corresponding expansion at Tp , with coordinate by , gives equation (4.3). A mask factor cannot vanish at distinct $p, q \in E$: $m_b(p - q) = 0$ would imply $Tp = Tq$, whereas T is a permutation of E . $\square$

Proposition 4.2 (Invariant-measure energy identity). *Every invariant probability measure μ satisfies*

$$\int \log A_{\mathbf{a}} \, d\mu = -DB + B \sum_i a_i \mu(\mathcal{O}_i). \quad (4.5)$$

This integral is finite even if $\int |\log \Phi_{\mathbf{a}}| \, d\mu = \infty$.

Proof. First suppose $\mu(E) = 0$. Invariance gives $\mu(T^{-n}E) = 0$ for every n . Thus $U = -\log \Phi_{\mathbf{a}}$ is finite almost everywhere, and the coboundary gives, almost everywhere,

$$U - U \circ T = \log A_{\mathbf{a}} + DB \leq DB.$$

By lemma 3.2, $\int \log A = -DB$. For the orbit measure δ_i , sum logarithms in equation (4.3) around the cycle. The $\log c_p$ terms telescope, leaving $(a_i - D)B$.

Finally decompose an invariant measure as

$$\mu = \theta_0 \mu_0 + \sum_i \theta_i \delta_i, \quad \theta_i = \mu(\mathcal{O}_i), \quad \mu_0(E) = 0. \quad (4.6)$$

The restricted pieces are invariant: the mass of $T^{-1}\mathcal{O}_i \setminus \mathcal{O}_i$ is zero. Finite linearity proves the formula. Zero coefficients in the decomposition are omitted. $\square$

4.2 Proof of pressure and equilibrium selection

Proof of theorem 2.1. Set

$$S = \sup_{\mu \in \mathcal{M}_T, \mu(E)=0} \left(h_\mu(T) + \int \log v \, d\mu \right).$$

This supremum need not be attained and can be $-\infty$. The decomposition above, affinity of entropy under finite invariant mixtures, and lemma 3.1 give

$$\mathcal{P}_b(v) = \max\{S, Q_1(v), \dots, Q_m(v)\}, \quad (4.7)$$

$$\mathcal{P}_b(vA) = -DB + \max\{S, Q_1(v) + a_1B, \dots, Q_m(v) + a_mB\}. \quad (4.8)$$

Because $a_iB > 0$, replacing S in the second maximum by $\mathcal{P}_b(v)$ does not change the answer. This proves equation (2.7).

If $C = \mathcal{P}_b(v)$, then $Q_i(v) < \mathcal{P}_b(v)$ for every finite $Q_i(v)$: indeed $Q_i(v) + a_iB \leq C$. Hence all background equilibrium measures avoid E . Equality in equation (4.8) requires every positive-mass component of equation (4.6) to attain its winning branch. A component avoiding E can do so only when $C = \mathcal{P}_b(v)$ and then must be a background equilibrium measure. An orbit component can do so exactly when $Q_i(v) + a_iB = C$. Conversely, every mixture of these measures attains equality. The pressure is finite, so existence follows from lemma 3.1. $\square$

4.3 Why complete cycles rather than arbitrary phase sets?

Proposition 4.3 (Finite balance criterion). *A finite positive atomic measure $\eta = \sum_p a(p)\delta_p$ satisfies $T_*\eta = \eta$ if and only if its support is a union of periodic orbits and its masses are constant on each orbit. These are precisely the weighted phase sets for which the direct multiplication of sine quotients gives equation (4.1) with the same weighted sine product in numerator and denominator.*

Proof. The support of such an invariant finite measure is forward invariant. Its finite functional graph consists of cycles with possible incoming trees. An incoming leaf with positive mass but no positive-mass preimage contradicts invariance. Removing leaves excludes all transient trees. On a cycle invariance forces equal consecutive masses. The converse is immediate.

In the sine identity, the phase divisor in the numerator is $T_*\eta$, whereas the proposed sine product at Tx has phase divisor η . Equality forces the same zeros and vanishing orders, hence the same masses. This proves necessity for this direct product identity. $\square$

This does not classify every possible cohomological representation of an unbalanced weight using a different auxiliary function.

5 A sampling theorem for products of real powers

The passage from pressure to moments is not automatic for an arbitrary continuous weight with zeros. Polynomial structure supplies a uniform comparison, including when the exponents are nonintegral.

Lemma 5.1 (Subharmonic sampling). *Let $P_1, \dots, P_q$ be nonzero complex polynomials and $t_j > 0$. Set $F(z) = \prod_j |P_j(z)|^{t_j}$ and $d = \sum_j t_j \deg P_j$. If $d \leq \kappa N$, then*

$$N \int_0^1 F(e^{2\pi i x}) \, dx \leq \sum_{k=0}^{N-1} \sup_{x \in [k/N, (k+1)/N]} F(e^{2\pi i x}) \leq 48e^{\kappa/4} N \int_0^1 F(e^{2\pi i x}) \, dx. \quad (5.1)$$

The constant is independent of the zeros and their separation.

Proof. The lower bound follows by integrating on each arc. For the upper bound, $\log F = \sum_j t_j \log |P_j|$ is subharmonic with the usual value $-\infty$ at zeros. Exponentiation shows that F is subharmonic; alternatively one can first regularize the logarithms and pass to the limit in the disk mean inequality.

Choose a maximum point z_k on every arc and put $\rho = 1/(4N)$. The disks $D(z_k, \rho)$ have overlap at most three. Indeed, the centers of disks containing a common point lie on a circle arc of length less than $2\pi/N$; such an arc meets at most three of the partition arcs, including their endpoints. For $N = 1, 2$ the same bound is immediate. All disks lie in the annulus $1 - \rho < |z| < 1 + \rho$. Consequently

$$\sum_k F(z_k) \leq \frac{3}{\pi \rho^2} \int_{1-\rho < |z| < 1+\rho} F(z) dA(z).$$

Write $M(R) = (2\pi)^{-1} \int_0^{2\pi} F(Re^{i\theta}) d\theta$. Subharmonicity gives $M(R) \leq M(1)$ for $R \leq 1$. For $R \geq 1$, use the reversed polynomials $P_j^*(z) = z^{\deg P_j} \overline{P_j(1/\bar{z})}$. Their product of absolute powers, denoted F^* , satisfies

$$F(Re^{i\theta}) = R^d F^*(R^{-1}e^{i\theta}), \quad M^*(1) = M(1).$$

Thus $M(R) \leq R^d M(1) \leq e^{\kappa/4} M(1)$ throughout the outer part of the annulus. Integrating in polar coordinates gives

$$\int_{1-\rho < |z| < 1+\rho} F(z) dA(z) \leq 4\pi \rho e^{\kappa/4} M(1).$$

The resulting bound is $12\rho^{-1}e^{\kappa/4}M(1)$, which is equation (5.1). No attempt is made to optimize the constant. $\square$

Corollary 5.2 (Moment–pressure conversion). *Suppose $w(x) = \prod_{j=1}^q |p_j(e^{2\pi i x})|^{t_j}$ with fixed nonzero polynomials p_j and fixed $t_j > 0$. If $\mathcal{P}_b(w)$ is finite, then*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \int_0^1 w^{(n)}(x) dx = \mathcal{P}_b(w) - B. \quad (5.2)$$

Proof. The depth- n polynomial corresponding to p_j is $\prod_{r=0}^{n-1} p_j(z^{b^r})$. Its degree is $\deg p_j(N-1)/(b-1)$. The weighted sum of these degrees is at most κN for a fixed κ . Apply lemma 5.1 and take logarithms divided by n . Both multiplicative comparison constants are independent of n . $\square$

A Laurent-polynomial factor can be multiplied by a monomial without changing its absolute value on the unit circle. The same conclusion therefore applies to that representation as well.

6 A squared background has exactly zero pressure

6.1 Digit factorization and Parseval

Unique base- b digit expansion gives the polynomial identity

$$\begin{aligned} q_{n,c}(z) &= \prod_{j=0}^{n-1} \left(\frac{1}{b} \sum_{r=0}^{b-1} e^{-2\pi i r c} z^{rb^j} \right) \\ &= \frac{1}{N} \sum_{k=0}^{N-1} e^{-2\pi i c s_b(k)} z^k. \end{aligned} \quad (6.1)$$

On the unit circle, $|q_{n,c}(e^{2\pi i x})| = \prod_{j=0}^{n-1} m_b(T^j x - c)$ and $S_{n,c}(x) = N q_{n,c}(e^{2\pi i x})$. All coefficients of $q_{n,c}$ have absolute value $1/N$, so Parseval gives

$$\int_0^1 |q_{n,c}(e^{2\pi i x})|^2 dx = \frac{1}{N}. \quad (6.2)$$

Proposition 6.1 (Phase-independent squared pressure). *For every real c and every integer base $b \geq 2$,*

$$\mathcal{P}_b(m_b(\cdot - c)^2) = 0. \quad (6.3)$$

Proof. The weighted degree of $|q_{n,c}|^2$ is $2(N - 1)$. Combining lemma 5.1 with equation (6.2) yields $1 \leq Z_n(m_b(\cdot - c)^2) \leq 48e^{1/2}$. Take n th logarithmic rates. The assertion is exact in c , including phases at which a selected periodic orbit meets a zero. $\square$

This argument depends on the square and on the equal coefficient magnitudes. It does not evaluate a general nonquadratic background pressure.

6.2 Proof of the global mixed-moment formula

Proof of theorem 2.3. Put $v_c(x) = m_b(x - c)^2$ and $w = v_c A_{\mathbf{a}}$. By theorem 2.1 and proposition 6.1,

$$\mathcal{P}_b(w) = -DB + \max\{0, \max_i(Q_i(c) + a_i B)\}.$$

All masks are absolute values of nonzero polynomials on the unit circle. Thus corollary 5.2 applies, and the factorization gives

$$I_n(\mathbf{a}, c) = N^{D+2} \int_0^1 w^{(n)}(x) dx.$$

Dividing the logarithm by $\log N = nB$ proves equation (2.12). Each finite moment is positive, since the integrand has only finitely many zeros. Real positive exponents cause no difficulty in either the sampling or variational argument. $\square$

6.3 One anchored fixed point

Take the fixed point 0 with exponent $a > 0$. Since $D = a$,

$$\mathcal{P}_b(m_b(\cdot)^a m_b(\cdot - c)^2) = \max\{-aB, 2 \log m_b(c)\}, \quad (6.4)$$

$$\lim_{n \rightarrow \infty} \frac{\log I_n(a, c)}{\log N} = 1 + \max\{0, a + 2 \log_b m_b(c)\}. \quad (6.5)$$

If $0 < m_b(c) < 1$, the two branches exchange dominance at

$$a_*(c) = -2 \log_b m_b(c). \quad (6.6)$$

If $m_b(c) = 0$, that orbit is excluded from equilibrium at every finite a , and no finite transition occurs. At $c = 0$, the threshold is zero and the exponent is $a + 1$, consistently with the $(a + 2)$ nd absolute moment of a Dirichlet kernel.

In base two and for $|c| < 1/2$,

$$a_*(c) = -\frac{2 \log \cos \pi c}{\log 2} = \frac{\pi^2}{\log 2} c^2 + \frac{\pi^4}{6 \log 2} c^4 + O(c^6). \quad (6.7)$$

This is a parabolic boundary for a mixed family, not the square-root phase response of the single critical mask in the neighboring repository manuscripts.

7 An explicit three-branch phase diagram

In base two, use $\mathcal{O}_0 = \{0\}$ with exponent a and $\mathcal{O}_2 = \{1/3, 2/3\}$ with exponent d . Then $D = a + 2d$, and

$$Q_0(c) = 2 \log |\cos \pi c|, \quad (7.1)$$

$$\begin{aligned} Q_2(c) &= \log |\cos \pi(1/3 - c) \cos \pi(2/3 - c)| \\ &= \log \left| \frac{1}{4} - \frac{1}{2} \cos 2\pi c \right|. \end{aligned} \quad (7.2)$$

At $c = 1/4$, $Q_0 = -B$ and $Q_2 = -2B$. Therefore

$$\boxed{\frac{\mathcal{P}_2(v_{1/4} A_{a,d})}{\log 2} = -a - 2d + \max\{0, a - 1, d - 2\}. \quad (7.3)}$$

Equivalently,

$$\lim_{n \rightarrow \infty} \frac{1}{\log N} \log \int_0^1 |S_{n,1/4}|^2 |S_{n,0}|^a |S_{n,1/3} S_{n,2/3}|^d dx = 1 + \max\{0, a - 1, d - 2\}. \quad (7.4)$$

The background region is $a < 1, d < 2$. The fixed-point region is $a - 1 > \max\{0, d - 2\}$; the period-two region is $d - 2 > \max\{0, a - 1\}$. At $(a, d) = (1, 2)$ the three branches coexist. The equilibrium measures there are exactly the mixtures of background equilibrium measures, δ_0 , and $(\delta_{1/3} + \delta_{2/3})/2$. This description does not assume that the background equilibrium measure is unique.

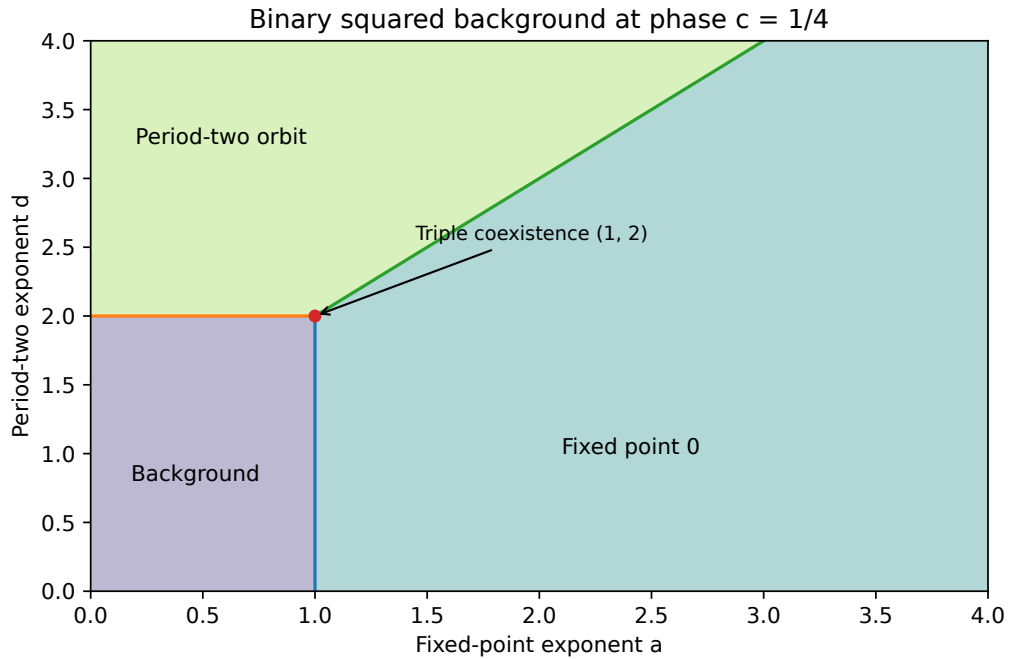


Figure 1: The mixed-moment exponent in equation (7.4). The three affine branches meet at $(1, 2)$. This is an exact phase diagram; the plot is only its visualization.

In the general vector of exponents, the background pressure branch has gradient $-B(r_1, \dots, r_m)$ and orbit branch j has gradient $B(e_j - (r_1, \dots, r_m))$. These differ. A path crossing an active boundary transversely therefore has a first-derivative jump. A path tangent to the boundary need not exhibit that jump in its restriction.

8 Leading asymptotics without a background

Pressure determines a logarithmic growth rate, not its amplitude. For the pure anchored product define

$$J_n(\mathbf{a}) = \int_0^1 \prod_i \prod_{p \in \mathcal{O}_i} |S_{n,p}(x)|^{a_i} dx. \quad (8.1)$$

Telescoping equation (4.1), with the powers of N from equation (6.1), gives the exact integral

$$J_n(\mathbf{a}) = \int_0^1 \frac{\Phi_{\mathbf{a}}(Nx)}{\Phi_{\mathbf{a}}(x)} dx. \quad (8.2)$$

The quotient has removable values at E : T^n permutes each orbit, and numerator and denominator have the same local vanishing order. Abbreviate $\Phi = \Phi_{\mathbf{a}}$, $\bar{\Phi} = \int_0^1 \Phi$, $a_* = \max_i a_i$, and $L = \text{lcm}(r_1, \dots, r_m)$. Recall the positive local constants c_p in equation (4.2).

Theorem 8.1 (Constants, logarithms, and periodic amplitudes). *The following three alternatives hold.*

(i) If $a_* < 1$, then

$$J_n \longrightarrow \bar{\Phi} \int_0^1 \Phi(x)^{-1} dx. \quad (8.3)$$

(ii) If $a_* = 1$, then

$$J_n = 2\bar{\Phi} \left(\sum_{p \in E: a(p)=1} \frac{1}{c_p} \right) \log N + O(1). \quad (8.4)$$

(iii) If $a_* > 1$, then

$$J_n = N^{a_*-1} (C_{n \bmod L} + o(1)), \quad (8.5)$$

where the positive, finite constants are

$$C_r = \sum_{p \in E: a(p)=a_*} \frac{1}{c_p} \int_{\mathbb{R}} \frac{\Phi(T^r p + u)}{|u|^{a_*}} du, \quad 0 \leq r < L. \quad (8.6)$$

The actual period can be a proper divisor of L .

Proof. Choose pairwise disjoint fixed neighborhoods of the points of E . Near $p \in \mathcal{O}_i$, $\Phi(p + y) = c_p |y|^{a_i} (1 + O(|y|))$. Constants can be chosen uniformly over the finite set of points.

For (i), $\Phi^{-1} \in L^1(\mathbb{T})$. If f is continuous and periodic and $g \in L^1(\mathbb{T})$, then

$$\int_0^1 f(Nx)g(x) dx \longrightarrow \left(\int_0^1 f \right) \left(\int_0^1 g \right) \quad (N \rightarrow \infty \text{ through integers}).$$

To see this, approximate f uniformly by trigonometric polynomials; for each nonconstant frequency use the vanishing at infinity of the Fourier coefficients of g , which itself follows by L^1 approximation by trigonometric polynomials. Taking $f = \Phi$ and $g = \Phi^{-1}$ proves the claim.

For (ii), neighborhoods of points with $a(p) < 1$ and the complement of all neighborhoods contribute $O(1)$. At a point with exponent one, replace $\Phi(p + y)^{-1}$ by $(c_p |y|)^{-1}$. The resulting error is $O(1)$, since the numerator is bounded. On $|y| < 1/N$ the original integral is $O(1)$ as well: its numerator is $\Phi(T^n p + Ny) = O(|Ny|)$, uniformly in the finitely many images of p . For $1/N < y < \delta$, change variables $u = Ny$ to obtain

$$\int_1^{N\delta} \frac{\Phi(T^n p + u)}{u} du = \bar{\Phi} \log N + O(1).$$

Indeed the periodic function $\Phi(q + u) - \bar{\Phi}$ has a bounded primitive, uniformly in q , and integration by parts bounds its integral against $1/u$. The negative side supplies the same leading term. Sum over the exponent-one points.

For (iii), restrict first to a subsequence with $n \equiv r \pmod{L}$. At a point with exponent a_* set $y = u/N$ and divide its contribution by N^{a_*-1} . The integrand converges to $\Phi(T^r p + u)/(c_p |u|^{a_*})$. For $|u| \leq 1$ the numerator vanishes to order a_* , so the quotient is bounded. For $|u| > 1$ it is bounded by a constant times $|u|^{-a_*}$, an integrable function. The local denominator estimate gives these bounds uniformly on $|u| < N\delta$. Dominated convergence proves the stated contribution.

A point of smaller exponent a_i contributes at most $O(1 + \log N + N^{\max\{a_i-1, 0\}})$ by splitting at $1/N$ and using boundedness outside the inner interval. This is negligible compared with N^{a_*-1} . The complement contributes $O(1)$. There are only finitely many residue classes, so the individual subsequential limits give equation (8.5). Each limiting integral is finite by the bounds just given and is positive because Φ is positive off a finite set. $\square$

For a single anchor 0 with exponent one, $\bar{\Phi} = 2/\pi$ and $c_0 = \pi$, giving the familiar coefficient $4/\pi^2$ of $\log N$. For exponent two the limiting coefficient is one, also consistent with the exact Parseval identity $J_n = N$. These one-anchor normalizations are consistency checks, not new claims about classical Dirichlet moments. The distinction specific to multiple periodic anchors is that the constants in equation (8.6) can really differ.

9 A uniform multivariate critical window

The logarithm at exponent one is a transition between an integrable singularity and concentration near an orbit. Several orbits can enter this window independently, and their leading contributions add rather than select a single maximum.

Fix a nonempty set C of orbit indices. Choose baseline exponents $a_i^0 = 1$ for $i \in C$ and $0 < a_i^0 < 1$ for $i \notin C$. For bounded real parameters u_i , $i \in C$, define

$$a_i(N) = 1 + \frac{u_i}{\log N} \quad (i \in C), \quad a_i(N) = a_i^0 \quad (i \notin C). \quad (9.1)$$

These exponents are positive for all sufficiently large N . Let $\Phi_0 = \Phi_{a^0}$, let c_p^0 be its local constants, and set

$$K_i = \sum_{p \in \mathcal{O}_i} \frac{1}{c_p^0}, \quad \Psi(u) = \begin{cases} (e^u - 1)/u, & u \neq 0, \\ 1, & u = 0. \end{cases} \quad (9.2)$$

Theorem 9.1 (Critical-window crossover). *For every fixed $R < \infty$, uniformly for $|u_i| \leq R$,*

$$J_n(\mathbf{a}(N)) = 2 \left(\int_0^1 \Phi_0 \right) \log N \sum_{i \in C} K_i \Psi(u_i) + O_R(1). \quad (9.3)$$

Proof. Write $\Phi_N = \Phi_{\mathbf{a}(N)}$ and $\bar{\Phi}_N = \int_0^1 \Phi_N$. On compact positive exponent intervals, $t^\alpha |\log t|$ is bounded for $0 \leq t \leq 1$. Differentiating each factor with respect to its exponent, or integrating this bound, shows

$$\|\Phi_N - \Phi_0\|_\infty = O_R((\log N)^{-1}), \quad \bar{\Phi}_N = \bar{\Phi}_0 + O_R((\log N)^{-1}).$$

The explicit formula for c_p likewise gives $c_p(N) = c_p^0 + O_R((\log N)^{-1})$, uniformly over $p \in E$. Local expansions of Φ_N and their relative $O(|y|)$ errors are uniform because the exponents stay in a compact positive interval.

Points not on a critical orbit, and the complement of all fixed neighborhoods, contribute $O_R(1)$, as their denominator exponents stay strictly below one. Near a critical point write $a = 1 + u/\log N$.

Replacing the denominator by $c_p(N)|y|^a$ changes the integral by $O_R(1)$: the error is bounded by a constant times $|y|^{1-a}$, which is uniformly integrable. On $|y| < 1/N$ the numerator vanishes to the same order a , and this inner contribution is $O_R(N^{a-1}) = O_R(1)$.

On the positive outer interval $[1/N, \delta]$, replace $\Phi_N(T^n p + Ny)$ by its mean. The zero-mean remainder, considered as a function of y , has a primitive bounded by C_R/N . Integration by parts against y^{-a} bounds its contribution by $O_R(N^{a-1}) = O_R(1)$. The remaining elementary integral is

$$\int_{1/N}^{\delta} y^{-1-u/\log N} dy = \log N \Psi(u) + O_R(1). \quad (9.4)$$

For $u \neq 0$ this follows by evaluating the antiderivative; the result extends uniformly across $u = 0$, either by Taylor expansion or by integrating e^{ut} on $0 \leq t \leq 1$. The negative interval gives the same leading term. Replacing $\bar{\Phi}_N/c_p(N)$ by $\bar{\Phi}_0/c_p^0$ changes a term of size $O_R(\log N)$ by only $O_R(1)$. Summing the two sides of every critical point proves equation (9.3). $\square$

At $u_i = 0$, equation (9.3) reduces to equation (8.4). The scale is $1/\log N = 1/(n \log b)$ in the exponents. This result concerns pure anchored products; a squared nontrivial background at a phase boundary may have a different prefactor mechanism.

10 An exact seventh-root moment with three amplitudes

The residue-class dependence in theorem 8.1 cannot in general be removed. The orbit $\{1/7, 2/7, 4/7\}$ under doubling gives a small example that is exactly solvable at every depth.

Theorem 10.1 (Exact period-three moment). *For $N = 2^n$ and every integer $n \geq 0$,*

$$\int_0^1 |S_{n,1/7}(x)S_{n,2/7}(x)S_{n,4/7}(x)|^2 dx = \begin{cases} 3N - 2, & n \equiv 0 \pmod{3}, \\ 4N - 2, & n \equiv 1 \pmod{3}, \\ 4N + 4, & n \equiv 2 \pmod{3}. \end{cases} \quad (10.1)$$

In particular, the moment divided by N has residue-class limits 3, 4, 4, and does not converge.

Proof. Let $\zeta = e^{2\pi i/7}$ and $\eta = \zeta + \zeta^2 + \zeta^4$. The complementary root sum is $\bar{\eta} = \zeta^3 + \zeta^5 + \zeta^6$, and direct multiplication gives

$$\eta + \bar{\eta} = -1, \quad \eta\bar{\eta} = 2, \quad \eta^2 + \eta + 2 = 0. \quad (10.2)$$

Indeed the product of the two sums has three terms equal to one and each of the six nontrivial seventh roots once, giving $3 - 1 = 2$. Define

$$R(z) = \prod_{p \in \{1,2,4\}} (z - \zeta^p) = z^3 - \eta z^2 + \bar{\eta} z - 1. \quad (10.3)$$

Squaring permutes its roots, so $R(z)$ divides $R(z^N)$ for $N = 2^n$. Since $\Phi(x) = 4^{-3}|R(e^{2\pi i x})|^2$ for the exponent-two orbit, equation (8.2) and Parseval identify the desired moment with the squared coefficient norm of the polynomial $R(z^N)/R(z)$.

Let $\bar{R}$ mean coefficientwise conjugation. The six nontrivial seventh roots give

$$R(z)\bar{R}(z) = 1 + z + \dots + z^6, \quad \frac{1}{R(z)} = \frac{(1-z)\bar{R}(z)}{1-z^7}.$$

Thus, as a formal power series, $1/R(z) = \sum_{k \geq 0} h_k z^k$ with period-seven coefficient sequence

$$(h_0, \dots, h_6) = (-1, -\bar{\eta}, 1, -\eta, -1, 0, 0). \quad (10.4)$$

For $0 \leq k < N$, the coefficients in the three consecutive length- N blocks of $R(z^N)/R(z)$ are

$$\begin{aligned} & -h_k, \\ & -h_{k+N} + \bar{\eta}h_k, \\ & -h_{k+2N} + \bar{\eta}h_{k+N} - \eta h_k. \end{aligned} \tag{10.5}$$

The final two entries are zero because the quotient has degree $3N - 3$; no coefficient after these blocks is nonzero. This includes $N = 1$, where the quotient is simply one.

For $r \in \{1, 2, 4\}$ let $F_r(k)$ be the sum of the three squared absolute values in equation (10.5), replacing shifts by N with shifts by r and reading subscripts modulo seven. Using

$$|x + y\eta|^2 = x^2 - xy + 2y^2 \quad (x, y \in \mathbb{Z}), \tag{10.6}$$

one obtains the following finite table entirely in integer arithmetic:

r	$F_r(0)$	$F_r(1)$	$F_r(2)$	$F_r(3)$	$F_r(4)$	$F_r(5)$	$F_r(6)$	Sum
1	1	5	4	4	5	1	1	21
2	3	3	4	8	4	3	3	28
4	7	7	3	3	2	3	3	28

By Parseval, the moment is $\sum_{k=0}^{N-1} F_r(k)$, where $r = N \bmod 7$. Write $N = 7q + r$. For $r = 1$, this sum is $21q + 1 = 3N - 2$; for $r = 2$, it is $28q + 6 = 4N - 2$; for $r = 4$, it is $28q + 20 = 4N + 4$. Finally $2^n \bmod 7$ cycles through 1, 2, 4 according to $n \bmod 3$, which proves equation (10.1). $\square$

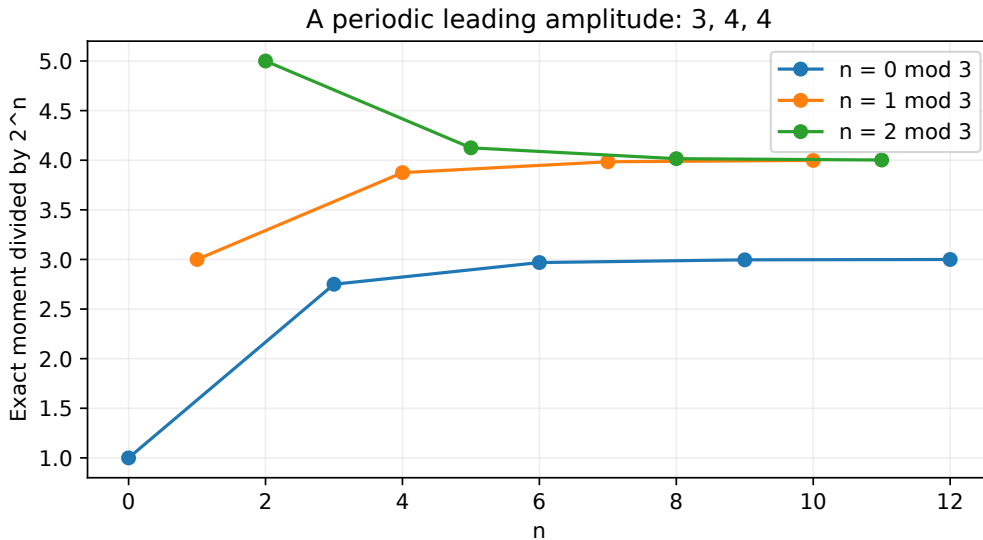


Figure 2: The exact seventh-root moment divided by 2^n . Its residue classes approach different limits. All plotted values come from the exact formula, not a numerical quadrature.

Here $D = 6$ and $a_* = 2$, so corollary 2.2 predicts $\mathcal{P}_2(A) = -4 \log 2$, and the unnormalized moment has exponent one. That correct exponent alone cannot distinguish the three leading amplitudes. Nor can their average replace $C_{n \bmod 3}$ with a single asymptotic equivalent.

11 Explicit realization of finite polyhedral pressure

The orbit rewards can also be used constructively. The following result prescribes a finite maximum of affine functions by an explicit finite product, while keeping the underlying dynamical map fixed.

Theorem 11.1 (Finite trigonometric realization). *Fix $b \geq 2$, a nonempty compact set $K \subset \mathbb{R}^d$, and real affine functions $\ell_1, \dots, \ell_m$ on $\mathbb{R}^d$. Choose m distinct periodic orbits $\mathcal{O}_i$ of T_b . There exist affine functions c and a_i , with $a_i(t) > 0$ for all $t \in K$, such that*

$$w_t(x) = e^{c(t)} \prod_i \prod_{p \in \mathcal{O}_i} m_b(x - p)^{a_i(t)} \quad \text{satisfies} \quad \mathcal{P}_b(w_t) = \max_i \ell_i(t) \quad (t \in K). \quad (11.1)$$

The equilibrium measures are exactly the convex combinations of δ_i for which $\ell_i(t) = \max_j \ell_j(t)$. For each prescribed finite integer $k \geq 0$, the construction can be made C^k in x , uniformly over $t \in K$.

Proof. Choose a constant $\ell_* < \min_{i,t \in K} \ell_i(t)$, and define

$$a_i(t) = 1 + \frac{\ell_i(t) - \ell_*}{B}, \quad D(t) = \sum_i r_i a_i(t), \quad c(t) = \ell_* + (D(t) - 1)B. \quad (11.2)$$

Every $a_i(t) > 1$. By corollary 2.2,

$$\begin{aligned} \mathcal{P}_b(w_t) &= c(t) - D(t)B + B \max\{1, a_1(t), \dots, a_m(t)\} \\ &= \max\{\ell_*, \ell_1(t), \dots, \ell_m(t)\} = \max_i \ell_i(t). \end{aligned}$$

The background branch is strictly smaller, so the equilibrium classification follows from theorem 2.1.

There are arbitrarily many distinct periodic orbits for fixed b . For example, $1/(b^r - 1)$ has exact period r whenever $b^r - 1 > 1$: a smaller period $j < r$ would require divisibility of $b^j - 1$ by $b^r - 1$, which is impossible. Choosing distinct such periods suffices.

Lowering ℓ_* further makes every exponent larger than k , uniformly on K . The zeros of m_b are simple in absolute-value coordinates, and $|y|^a$ is C^k for $a > k$. At any coincident mask zeros the exponents add, preserving this property. Away from zeros the factors are smooth. Compactness of the parameter set and a uniform positive margin above k give uniform continuity and bounds for the derivatives through order k . $\square$

This is a concrete realization result, not an assertion of the first pressure-flexibility theorem; see [4]. It also does not contradict analyticity theorems for strictly positive Hölder weights. Our weights vanish, so their logarithms take the value $-\infty$ at zeros. Moreover, the parameter family is not required to be of the restricted form $\exp(tf)$ for one everywhere-finite potential f . The smoothness assertion concerns w_t itself, not $\log w_t$.

12 A nonlinear expanding-map extension

The selection mechanism is not inherently trigonometric. Let $F: \mathbb{T} \rightarrow \mathbb{T}$ be a C^1 expanding local diffeomorphism and let $\mathcal{O}_i$ be finitely many disjoint periodic orbits of lengths r_i . Put

$$\chi_i = \frac{1}{r_i} \log |(F^{r_i})'(p)| > 0 \quad (p \in \mathcal{O}_i).$$

Choose a continuous $H \geq 0$ whose zero set is exactly $E = \bigcup_i \mathcal{O}_i$ and whose local form is

$$H(p + y) = c_p |y|^{a_i} (1 + o(1)), \quad c_p > 0, \quad p \in \mathcal{O}_i, \quad (12.1)$$

with $a_i > 0$. Such H exists, for example by a finite product of powers of distances with locally smooth positive factors. The ratio $R = H \circ F / H$ on $\mathbb{T} \setminus E$ extends continuously to a nonnegative weight, with

$$R(p) = |F'(p)|^{a_i} \frac{c_{F(p)}}{c_p} > 0. \quad (12.2)$$

It can vanish at points of $F^{-1}E \setminus E$.

Theorem 12.1 (Lyapunov-weighted periodic selection). *For every continuous $v \geq 0$ with finite pressure and every real constant K_0 ,*

$$\mathcal{P}_F(e^{-K_0}vR) = -K_0 + \max \left\{ \mathcal{P}_F(v), \max_i (Q_i(v) + a_i\chi_i) \right\}, \quad (12.3)$$

where $Q_i(v)$ is the uniform orbit average of $\log v$. The equilibrium measures are the corresponding convex combinations of background equilibrium measures and maximizing orbit measures, with the background included only when its branch is maximal.

Proof. Local differentiability of F and equation (12.1) give equation (12.2), and hence continuity at every zero of H . Away from E continuity is immediate, so R is bounded. For an invariant measure giving E zero mass, $F^{-1}E$ also has measure zero. After multiplying H by a positive constant we may assume $H \leq 1$. Then $U = -\log H$ is finite and nonnegative almost everywhere and $U - U \circ F = \log R$ is bounded above. The stationary cancellation lemma gives $\int \log R = 0$. On $\mathcal{O}_i$, the c_p factors telescope and $\int \log R d\delta_i = a_i\chi_i$.

The variational argument for weights with zeros is the same monotone approximation as in lemma 3.1. Expanding circle maps have upper semicontinuous entropy and the continuous-potential variational principle. Decompose invariant measures into their selected periodic components and the component avoiding E . The latter branch is unchanged by R , while orbit i gains $a_i\chi_i > 0$. The maximum and equality cases now follow exactly as in theorem 2.1; the scalar contributes $-K_0$. $\square$

Taking $F = T_b$, $H = \Phi_a$, and $K_0 = DB$ recovers the periodic-anchor theorem. The nonlinear extension is an exact pressure statement. We do not assert a polynomial digital-moment representation for a general nonlinear map.

13 Verification, dependencies, and formalization

13.1 What the executable checks establish

The companion `verify.py` uses exact integer pairs (x, y) to represent $x + y\eta \in \mathbb{Z}[\eta]$. Its multiplication rule is

$$(x, y)(u, v) = (xu - 2yv, xv + yu - yv),$$

and its norm is equation (10.6). The script recomputes the entire seven-column table in the proof of theorem 10.1. It also performs independent exact polynomial divisions of $R(z^{2^n})$ by $R(z)$ for $n = 0, \dots, 12$, checks zero remainders, and compares the squared coefficient norms with equation (10.1). The largest checked N is 4096. All exact assertions passed. These finite checks are valuable implementation tests; the all- n proof is the periodic coefficient argument, not extrapolation from the checks.

Separate floating-point diagnostics test the orbit coboundary in bases two and three and its average on additional periodic orbits. The mixed-moment diagnostic uses a midpoint grid with $64N$ samples at depths $n = 6, 8, 10, 12$. For the two-anchor example at $c = 1/4$, its last two depths give the following effective exponents:

a	d	Proved exponent	Effective exponent, $n = 10$ to 12
0.4	0.7	1	1.001264
3.0	0.5	3	3.000860
0.3	4.0	3	3.002809
1.0	2.0	1	1.134007

Here the effective exponent is $(\log I_{12} - \log I_{10})/(2 \log 2)$ using the computed moments. The last row is the triple point and converges much more slowly in these finite-depth diagnostics. The exact

exponent theorem permits subpower corrections, including logarithms; we do not interpret that row as a proven logarithmic asymptotic for the mixed family. The critical-window computation is likewise only a diagnostic of the bounded remainder asserted by [theorem 9.1](#).

The full numerical data are retained in `verification_results.json`. The execution summary is in `verification_run.txt`. The numerical integrations are not rigorous interval enclosures. The script does not certify an infinite-dimensional spectrum, and none of the proofs relies on a collocation eigenvalue.

13.2 Proof-dependency audit

Result	Necessary ingredients
Exact pressure and equilibrium faces	Continuous-potential variational principle; upper semicontinuity and affinity of entropy; monotone approximation at zeros; one-sided stationary cancellation; orbit permutation.
Global mixed-moment exponent	The pressure theorem; finite digit factorization; Parseval; the subharmonic sampling lemma.
Pure leading amplitudes	Exact telescoping; local vanishing orders; periodic averaging; bounded primitives; dominated convergence.
Critical-window formula	Uniform local expansions; an elementary power integral; the same bounded-primitive averaging argument.
Seventh-root exact formula	A quadratic integer ring; a cubic polynomial; seven-periodic reciprocal coefficients; Parseval.
Polyhedral realization	Affine normalization of the exact pressure formula; local regularity of absolute powers.
Nonlinear selection	Local differentiation at cycles and the same stationary and variational arguments.

No unreviewed critical-pressure asymptotic from ProveIt is an assumption of these results. Conversely, this article does not formally certify those neighboring manuscripts.

13.3 A realistic Lean decomposition

The finite algebra is a natural first tranche. Formalize the digit factorization, the permutation of a finite periodic orbit, the polynomial divisibility $R \mid R(z^{2^n})$, and the norm computation in $\mathbb{Z}[\eta]$. The exact seventh-root theorem can be split into a finite algebra lemma and a finite Fourier orthogonality statement. This route does not need thermodynamic formalism.

For the pressure theorem, formalize the stationary cancellation lemma with truncations first. Its only analytic inputs are invariance of the integral and monotone convergence. Next prove the invariant-measure energy formula, finite invariant decompositions, and the abstract maximum-of-branches lemma. One may initially state the latter for variational pressure, keeping the topological-pressure interface as a separate obligation. The final interface requires the expanding-map entropy facts and the zero-weight approximation.

Sampling and critical-window estimates form a later analytic tranche. They need subharmonic mean inequalities, approximation in L^1 , local domination, and quantified uniform remainders. No Lean source is claimed or included in the present package. The theorem decomposition is a roadmap, not an assertion that these formal obligations have already been discharged.

14 Further research questions and concrete targets

The following questions are deliberately narrower than asking for a general theory of all singular phase potentials. Each has a specific model, a diagnostic, or a proof interface supplied by this article.

Research question 1 (Breaking orbit balance). What replaces the exact maximum law when the anchor phases cease to form complete cycles, or exponents within a cycle become unequal?

A first target is the two-anchor family of section 7: displace one of $1/3, 2/3$ independently near $(a, d, c) = (1, 2, 1/4)$. Determine whether the three-branch corner persists, rounds, or splits, and find its leading local normal form and equilibrium selection. The exact balanced formula supplies the limiting branches but does not control the perturbation. The single-phase results in the repository do not by themselves settle this mixed perturbation.

Research question 2 (Mixed critical prefactors). At a boundary $a_i B + Q_i(c) = 0$, what is the asymptotic equivalent of the squared-background mixed moment after removing its proved power of N ?

The explicit triple point $c = 1/4, a = 1, d = 2$ is a useful benchmark. Prove whether a logarithm, a higher power of a logarithm, a periodic amplitude, or a different subpower correction occurs. The finite-depth row in section 13 is evidence only of slow approach, not a diagnosis. The pure-background-free theorem cannot simply be applied because the squared background introduces a new depth-dependent factor.

Research question 3 (When do the residue-class constants coincide?). Characterize orbit sets and exponents for which all C_r in equation (8.6) are equal.

The seventh-root example proves that equality is not automatic even for a single orbit and one even exponent. Spatial symmetries can force equality by transporting both the local constants and the integrals; determine useful necessary conditions and classify small periods. For even integer exponents, exact algebraic coefficient methods can provide a finite test in examples where the corresponding rational function is manageable.

Research question 4 (An algorithm for cyclotomic moment formulas). For finite rational periodic phase sets with positive even integer exponents, can every pure moment be expressed as a finite sum of powers of N with coefficients periodic in the depth?

The reciprocal of a polynomial with roots of unity has coefficients which are polynomial in the coefficient index on residue classes, with degree controlled by root multiplicity. Combine this fact with the divisible numerator at $N = b^n$, then sum squared coefficients on blocks. A full algorithm needs a proof of its block decomposition, its period, and its complexity; the cubic seventh-root computation is a small verified prototype rather than that general theorem.

Research question 5 (Infinitely many periodic anchors). Under which summability and separation conditions does a countable product of anchor weights define a nontrivial continuous weight, and does its pressure admit a corresponding supremum-of-orbit formula?

Finite truncations give exact formulas, but an infinite product can collapse to zero, and dense zeros can make continuity fatal to a nontrivial limit. Control uniform convergence, the singular transfer function, and escape of equilibrium mass before passing to a limit. These are genuine additional issues, not formal substitutions of a supremum for a maximum.

Research question 6 (Other background orders). Can the anchored reduction help determine moment exponents for a background $m_b(\cdot - c)^s$ beyond the exactly solved square?

The pressure theorem already reduces the answer to one single-phase quantity $\mathcal{P}_b(m_b(\cdot - c)^s)$ and finitely many orbit averages. Insert a separately justified formula in any regime where one is available. In particular, repository asymptotics can be combined with this theorem after their precise hypotheses and proof status are checked. A global exact expression for the single-phase background is not supplied here.

Research question 7 (Positive regularization and selection at coexistence). For $w_\varepsilon = w + \varepsilon$, determine the rate at which the smooth positive-weight pressure approaches an anchored corner and the limiting proportions of its equilibrium measures.

The proof of lemma 3.1 already gives monotone pressure convergence, but neither a rate nor a selected mixture. Competing periodic components and the background have different neighborhoods of zeros. An analysis of these neighborhoods could produce a finite-dimensional crossover law, but the exact maximum alone does not identify it.

Research question 8 (Certified analysis and formal proof). What is the smallest reusable formal interface that verifies the pressure reduction, and can the noninteger constants C_r be enclosed rigorously and efficiently?

For formal proof, separate invariant-measure optimization from the entropy and coding interface as in section 13. For numerical certification, split each integral for C_r into a small removable-singularity interval, a compact middle region, and tails bounded by $|u|^{-a_*}$. Interval arithmetic on the compact region, with proven local and tail bounds, would turn the current diagnostics into certificates for concrete noninteger examples.

15 Conclusion

Complete periodic cycles make a mixed digital-product family exactly tractable. The singular coboundary does not change all invariant energies equally: it leaves an explicit periodic reward after the common shift. This distinction gives the global pressure formula and its full equilibrium faces. Parseval for a squared background and a real-exponent sampling estimate then produce explicit mixed-moment growth exponents, including exact phase boundaries and coexistence.

The leading asymptotic problem contains additional information. Without a background it has three regimes: a finite averaging constant, a critical logarithm, and periodic supercritical amplitudes. The seventh-root formula shows that the arithmetic oscillation is real, not a defect of a proof technique. The critical-window theorem and the finite pressure-realization construction provide further concrete consequences of the same mechanism.

These results solve the stated structured mixed problems. They do not settle global regularity for arbitrary phase tuples, establish a universal mixed critical prefactor, or constitute a Lean verification. Their strongest claims are the precise formulas and proofs above; independent review and a broader priority search remain appropriate before describing them as an established breakthrough.

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